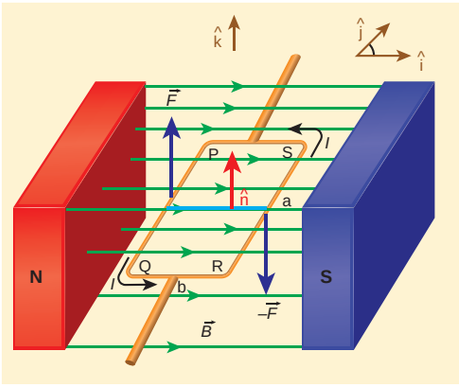
3.11.1 Expression for torque on a current loop placed in a magnetic field



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Figure 3.62 Rectangular coil placed in a magnetic field

Consider a single rectangular loop PQRS kept in a uniform magnetic field $\vec{B}$. Let a and b be the length and breadth of the rectangular loop respectively. Let $\hat{n}$ be the unit vector normal to the plane of the current loop. This unit vector $\hat{n}$ completely describes the orientation of the loop. Let $\vec{B}$ be directed from north pole to south pole of the magnet as shown in Figure 3.62.

When an electric current is sent through the loop, the net force acting is zero but there will be net torque acting on it. For the sake of understanding, we shall consider two configurations of the loop; (i) unit vector $\hat{n}$ pointing perpendicular to the field (ii) unit vector pointing at an angle θ with the field.

(i) when unit vector $\hat{n}$ is perpendicular to the field

In the simple configuration, the unit vector $\hat{n}$ is perpendicular to the field and plane of the loop is lying on xy plane as shown in Figure 3.62. Let the loop be divided into four sections PQ, QR, RS and SP. The Lorentz force on each loop can be calculated as follows:

(a) Force on section PQ

For section PQ, $\vec{l} = -a\,\hat{j}$ and $\vec{B} = B\,\hat{i}$

$$\vec{F}_{PQ} = I\vec{l} \times \vec{B}$$
$$= -IaB(\hat{j} \times \hat{i}) = IaB\,\hat{k}$$

Since the unit vector normal to the plane $\hat{n}$ is along the direction of $\hat{k}$.

(b) The force on section QR

$$\vec{l} = b\vec{i} \text{ and } \vec{B} = B\,\hat{i}$$

$$\vec{F}_{QR} = I\vec{l} \times \vec{B} = IbB(\hat{i} \times \hat{i}) = \vec{0}$$

(c) The force on section RS

$$\vec{l} = a\,\hat{j} \text{ and } \vec{B} = B\,\hat{i}$$

$$\vec{F}_{RS} = I\vec{l} \times \vec{B} = IaB(\hat{j} \times \hat{i}) = -IaB\,\hat{k}$$

Since, the unit vector normal to the plane is along the direction of $-\hat{k}$.

(d) The force on section SP

$$\vec{l} = -b\,\hat{i} \text{ and } \vec{B} = B\,\hat{i}$$

$$\vec{F}_{SP} = I\vec{l} \times \vec{B} = -IbB(\hat{i} \times \hat{i}) = \vec{0}$$

The net force on the rectangular loop is

$$\vec{F}_{net} = \vec{F}_{PQ} + \vec{F}_{QR} + \vec{F}_{RS} + \vec{F}_{SP}$$
$$\vec{F}_{net} = IaB\,\hat{k} + \vec{0} - IaB\,\hat{k} + \vec{0} \Rightarrow \vec{F}_{net} = \vec{0}$$

Hence, the net force on the rectangular loop in this configuration is zero. Now let us calculate the net torque due to these forces about an axis passing through the center

$$\vec{\tau}_{net} = \sum_{i=1}^{4} \vec{\tau}_i = \sum_{i=1}^{4} \vec{r}_i \times \vec{F}_i$$
$$= \left(\frac{b}{2}IaB + 0 + \frac{b}{2}IaB + 0\right)\hat{j}$$
$$\vec{\tau}_{net} = abIB\,\hat{j}$$

Since, A = ab is the area of the rectangular loop PQRS, the net torque for this configuration is

$$\vec{\tau}_{net} = ABI\,\hat{j}$$

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